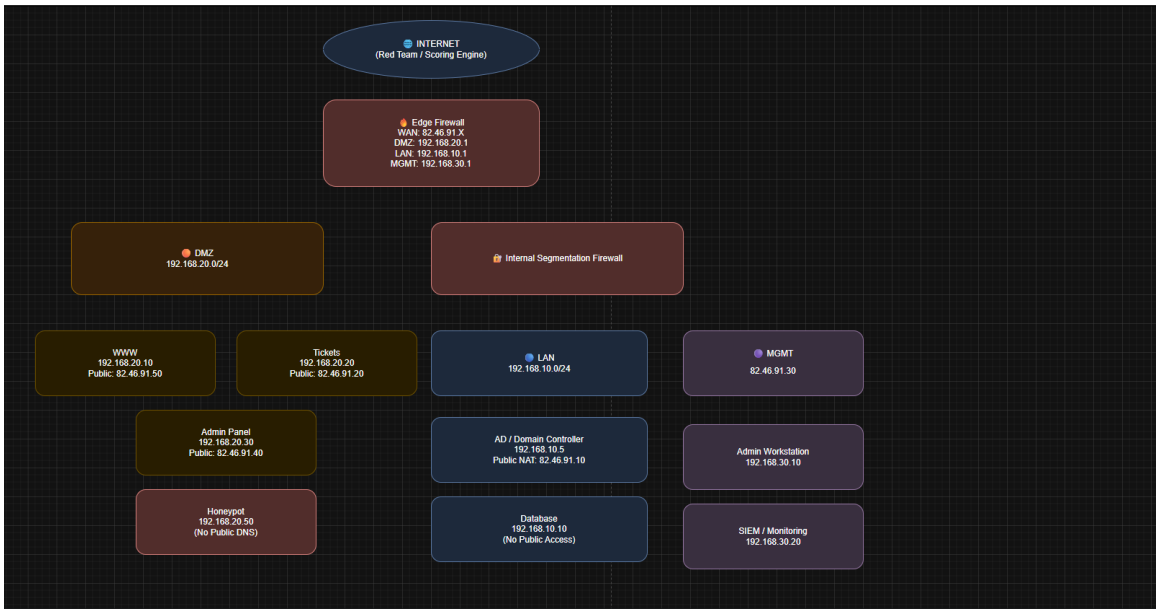
Team 31 - Service Status & Network Documentation

Team Name: GazulliShield

Environment: CDC / Blue Team Infrastructure

1. Network Overview

Describe the overall topology (Internet -> Firewall -> DMZ -> LAN -> MGMT).



2. Service Status Table

Service Name	URL / FQDN	Port	Server IP	Protocol	Purpose
AD LDAP	ad.team31.isucdc.com	389	**82.46.91.10**	TCP	Active Directory LDAP
AD RDP	ad.team31.isucdc.com	3389	_____	TCP	Remote Desktop to AD
AD LDAPS	ad.team31.isucdc.com	636	_____	TCP	Secure LDAP
TICKETS HTTP	tickets.team31.isucdc.com	5000	**82.46.91.20**	HTTP	Ticket API Health
TICKETS RDP	tickets.team31.isucdc.com	3389	_____	TCP	RDP to Ticket Server
MGMT HTTP	mgmt.team31.isucdc.com	5000	82.46.91.30	HTTP	Management Routes
MGMT SSH	mgmt.team31.isucdc.com	22	_____	SSH	Admin SSH Access
ADMIN HTTP	admin.team31.isucdc.com	80	82.46.91.40	HTTP	Admin Panel
ADMIN RDP	admin.team31.isucdc.com	3389	_____	TCP	Admin RDP
WWW HTTP	www.team31.isucdc.com	80	82.46.91.50	HTTP	Public Website
WWW SSH	www.team31.isucdc.com	22	_____	SSH	Web Server SSH
ADMIN Login	admin.team31.isucdc.com/login	80	_____	HTTP	Admin Authentication
WWW Login	www.team31.isucdc.com/login	80	_____	HTTP	User Authentication

3. IP Addressing Plan

Fill the table below with actual static IP addresses and network zones.

Server Name	IP Address
AD	82.46.91.10
TICKETS HTTP	82.46.91.20
MGMT HTTP	82.46.91.30
ADMIN HTTP	82.46.91.40
WWW HTTP	82.46.91.50

4. Firewall Configuration

Document the following:

- NAT Rules (WAN -> Internal IP + Port)

- Inbound Rules

- Outbound Rules

- Blocked Ports

- VLAN Segmentation

5. Security Hardening Measures

Include the following:

1. Management Backend (mgmt.team31.isucdc.com – AlmaLinux 8.4)

Service Review

- Verified `backend` systemd service configuration.
- Ensured application runs under a non-root service account.
- Confirmed required HTTP (port 5000) and SSH (port 22) availability.

Environment Security

- Secured `/etc/backend.env`:
  - Ownership set to `root`
  - Restricted permissions to prevent unauthorized read access
- Removed debug configuration (Werkzeug debug PIN).
- Ensured Flask runs in production mode (no debug/reload).

Database Review

- Audited MariaDB users.
- Confirmed no anonymous accounts.
- Verified root accounts restricted to local access.
- Ensured no unnecessary remote DB exposure.

2. Public Website (www.team31.isucdc.com – AlmaLinux 9)

SSH Hardening

- Reviewed `/etc/ssh/sshd_config` and included config files.
- Confirmed:
  - `PasswordAuthentication yes`
  - `PermitRootLogin` appropriately configured
  - PAM enabled
- Restricted SSH access using `AllowUsers`.
- Verified required administrative access remains functional.
- Confirmed no unintended account lockouts.

User Management

- Created dedicated administrative account (`UJIITHARTE`) with sudo privileges.
- Verified `/bin/bash` login shell.
- Audited faillock and account expiration status.
- Ensured default required users retain access per competition rules.

3. Ticketing & Service Stability

Service Monitoring

- Investigated intermittent HTTP failures on port 5000.
- Verified scheduled task operation.
- Confirmed service restarts and dependency connectivity (AD + backend).
- Ensured required health endpoints remain accessible.

4. General Hardening Measures

- Verified all required ports remain accessible per scenario requirements.
- Avoided blocking competition network access.
- Maintained required functionality for:
  - AD authentication
  - Backend API
  - Admin UI
  - Public Website
- Confirmed no unnecessary services exposed.
- Ensured administrative accounts maintain realistic privileges.

5. Compliance Confirmation

All required services:

- Remain accessible from the competition network
- Preserve original functionality
- Maintain administrative access as required
- Follow principle of least privilege where applicable

6. Monitoring & Logging

Document monitoring tools used:

- Suricata

- SIEM (ELK / Splunk)

- Log forwarding configuration

- Backup configuration